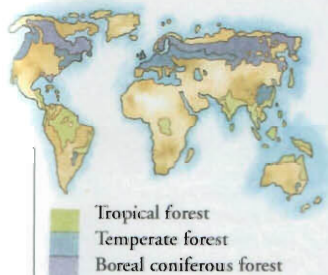


FORESTS



A THIRD OF THE WORLD'S land surface consists of forest – areas of land covered by dense tree cover. Each forest is an ecosystem – a group of animals and plants interacting with the physical environment and one another. More plants and animals live in forests than in any other environment. Forests differ according to the climate – boreal, temperate, or tropical. They help maintain the Earth's natural balance; trees absorb and release gases which regulate the climate.



Boreal coniferous forest

Down-angled branches allow the trees to shed the weight of the snow in cold climates without breaking them.

Forest floor is dark with little plant growth, but vegetation decay enriches the soil.

Soil beneath these trees is acid and infertile.

Temperate forest



Understorey contains shrubs and young growth.

Tropical forest

Tropical forests, or rainforests, tend to thrive in warm and wet climates. The porous soil is generally rich in aluminium and iron. In one hectare there may be more than 200 species of trees, hundreds of birds, mammals, and reptiles, and thousands of insects.

Emergent trees rise up to 60 m (197 ft).

Canopy of trees 15 m (50 ft) above the ground

Plants twine around branches.



Tropical forest

Each forest has layers of vegetation. In a tropical forest emergent trees poke through the top. Below are the canopy, the understorey, and the forest floor.

Broad leaves of deciduous trees grow rapidly in summer, and are shed in winter.

Temperate forest

These forests are found in mild or temperate climates, where winters are cool and summers are warm. The majority of trees are deciduous, such as oak and beech. Many temperate forests have been cleared for farmland because the soil beneath is very fertile.

Boreal coniferous forest

In cool, northern, or boreal regions, such as North Asia, there are vast areas of boreal forests, sometimes known as taiga. These dense forests contain hardy coniferous or evergreen trees, such as spruce, pine, fir, and larch trees.

Deforestation

Each year, forest land the size of Washington State, USA is destroyed. Trees are cut down for farmland or timber. Deforestation can cause huge environmental problems, disturbing the soil and forest life. Fewer trees to absorb carbon dioxide may also disrupt the climate.

Cattle ranching

Huge areas of rainforest in Brazil are cut down for cattle-ranching, which exhausts the soil within a few years.



Slash and burn

Nomadic farmers slash and burn forests for farmland. After a few years, they move on to allow the soil to regenerate.



Overgrazing

A shortage of land forces nomadic farmers to stay in one place. The land is overgrazed, and the soil dries up.



Afforestation

More people are becoming aware of the value of forests. In Southeast Asia, new forest land is created with tree-planting programmes. Some forests are now conservation areas.



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